

Erdős Problem #415

EP-415

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

For any n let $F(n)$ be the largest k such that any of the $k!$ possible ordering patterns appears in some sequence of $\phi(m+1), \dots, \phi(m+k)$ with $m+k \leq n$. Is it true that $F(n) = (c + o(1)) \log \log \log n$ for some constant c ? Is the first pattern which fails to appear always $\phi(m+1) > \phi(m+2) > \dots \phi(m+k)$? Is it true that 'natural' ordering which mimics what happens to $\phi(1), \dots, \phi(k)$ is the most likely to appear?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>